Auditing Carbon-Aware AI Model Selection

Accuracy, Sustainability, Transparency and Robustness in German Power-System Analytics

AI, Power & Responsibility - Governing Intelligent Systems for Sustainability

01 MOTIVATION

- Accuracy-only ML model choice hides compute and carbon costs.
- German grid analytics needs robust, season-aware evidence.
- Transparent reporting should include F1, AUC, runtime and size.
- Carbon-aware scheduling can reduce avoidable training impact.

02 DATASETS

OPSD Time Series
~50,401 raw hourly rows
load, wind, solar, TSO zones

Processed ML Table
50,209 rows; 29 features
lags, residual-load proxy

Artifacts
5 notebooks; 10 figures
CSV tables + IEEE report

03 METHODOLOGY PIPELINE

Data collection
OPSD hourly data

Preprocessing
interpolate, align UTC

Feature engineering
lags, RES, TSO zones

Modeling
LR, SVM, RF, HGB, NN

Evaluation
F1, AUC, runtime

Insights
Pareto + guidance

04 RESEARCH QUESTIONS

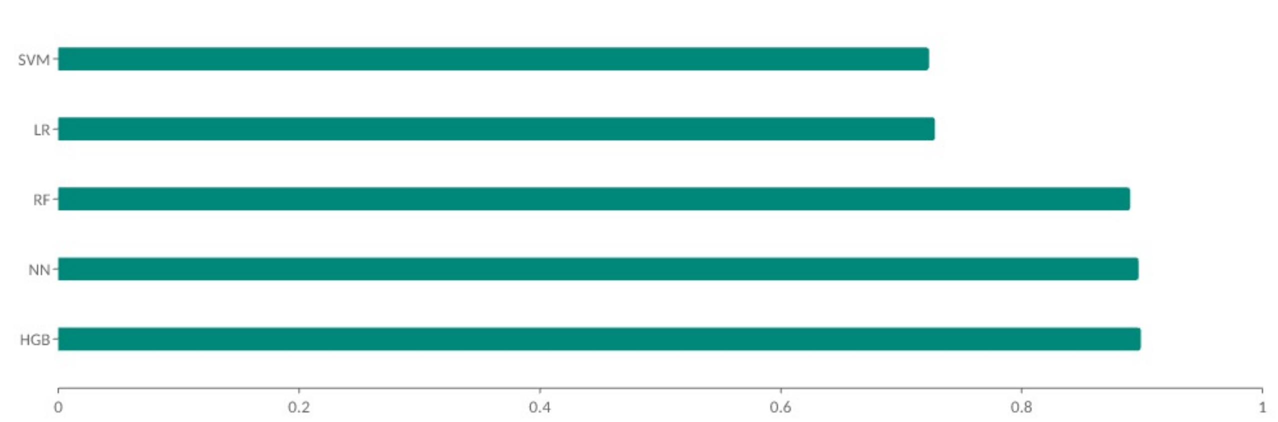
1. Which model is Pareto-efficient?
2. Which feature families add value?
3. Are rankings seasonally stable?
4. Can simple models explain drivers?
5. Do green training windows help?

06 KEY FINDINGS

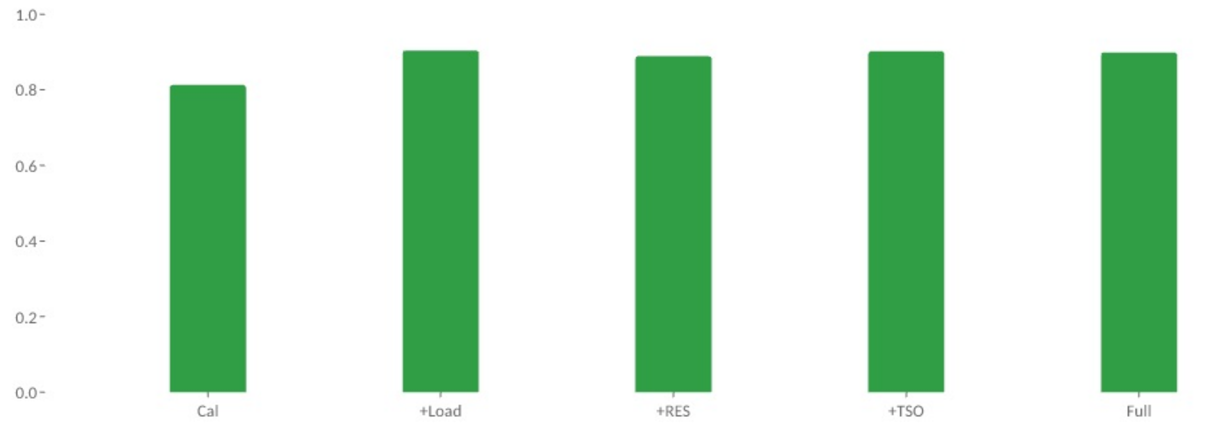
- HGB: best F1 and Pareto-efficient.
- Load history gives the largest feature gain.
- Summer is the hardest seasonal regime.
- Use low-compute baselines for auditability.

05 KEY RESULTS (HIGHLIGHTS)

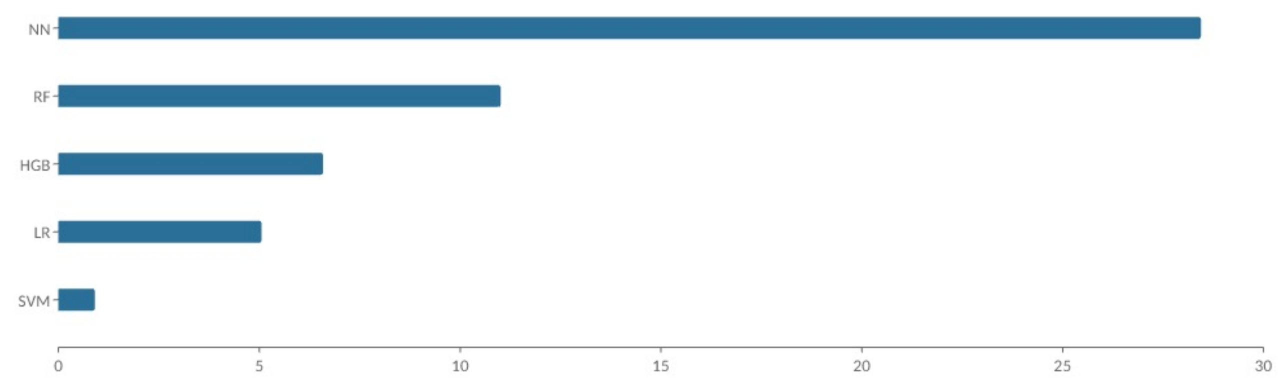
F1 by model



Feature ablation F1



Carbon proxy (mg CO2e)



31.9% green-window reduction

0.899 best F1

0.993 ROC-AUC

HGB is the balanced recommendation; Linear SVM is the lowest-compute fallback.

07 CONCLUSION

Responsible model selection should be Pareto-based: accuracy, carbon proxy, runtime, model size, seasonal robustness and auditability are evaluated together.

Recommended balanced model: Histogram Gradient Boosting. Recommended low-compute fallback: Linear SVM. Next step: replace proxy energy with measured hardware power.

Dataset: Open Power System Data Time Series (2020-10-06) | Reproducible notebooks, figures, tables, workflow, and IEEE/PES report